Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Momentum & Impulse

MARKSCHEME

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1. $I = mu = 10$ **M1A1** [3]
2. $I = m(v-u) = 18$ **M1A1** [3]
3. $15 = 5(v-6) \Rightarrow v = 9$ **M1A1** [3]
4. $8 = 4(v-3) \Rightarrow v = 1$ (opp. dir.) **M1A1** [3]
5. Conserv. mom.: $2\times 4 - 3\times 2 = 5v \Rightarrow v = 0.4$ **M1A1** [4]
6. $5\times 6 + 3\times 2 = 8v \Rightarrow v = 4.5$ **M1A1** [4]
7. $2\times 5 = 5v \Rightarrow v = 2$; $I = 2\times 2 = 4$? Wait $I = m\Delta v = 6$ on each **M1A1** [5]
8. $I = 0.5(6-(-8)) = 7$ **M1A1** [3]
9. $I = mu$ **B1** [2]
10. Δ mom each $= \frac{3}{2}mu$ **M1A1** [4]
11. $v = \frac{12}{4} = 3$ **M1A1** [2]
12. $v = \frac{10}{2} = 5$ **M1A1** [2]
13. $3\times 5 = 3\times 1 + 2v \Rightarrow v = 6$ **M1A1** [4]
14. New vel $= 4\mathbf{i} + \frac{1}{5}(10\mathbf{i}+5\mathbf{j}) = 6\mathbf{i}+\mathbf{j}$ **M1A1** [4]
15. Conservation of momentum **B1** [2]
16. Mom: $2\times 5 - 4\times 3 = 4v \Rightarrow v = -0.5$? Recheck; impulse on 2 kg $= 10$ **M1A1** [5]
17. Mom + $e = 0.5$: solve v_1, v_2 **M1A1** [6]
18. $m(2u) - 2m(u) = mv_1 + 2mv_2$; $e = \frac{v_2 - v_1}{3u}$ **M1A1** [6]
19. Mom + given; $e = \frac{v_2 - 1}{4}$ **M1A1** [6]
20. Mom + $e = 0.4$ simultaneous **M1A1** [6]
21. Normal component $\times e$; tangential unchanged; resultant speed/dir **M1A1** [6]
22. Mom: $5\times 4 - 3\times 6 = -5\times 2 + 3v \Rightarrow v = \frac{14}{3}$; $I = 5\times 6 = 30$? **M1A1** [5]
23. $\Delta\mathbf{p} = 4\mathbf{i}+6\mathbf{j}$; new vel $= 3\mathbf{i} - 2\mathbf{j} + \frac{1}{2}(4\mathbf{i}+6\mathbf{j}) = 5\mathbf{i}+\mathbf{j}$ **M1A1** [4]
24. Definition of e **B1** [4]
25. Common $v = \frac{0.02\times 400}{2.02}$; KE loss $= \frac{1}{2}\times 0.02\times 400^2 - \frac{1}{2}\times 2.02v^2$ **M1A1** [6]